Ethan W. Todd

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Education

University of Michigan — **Ann Arbor, MI**
Candidate, Ph.D. Physics — *2023 (Fall) – Present (expected 2026 (Fall))*
Advisors: Prof. Igor Jovanovic, Prof. James Wells
2025 (Winter) M.S. Physics

Florida State University — **Tallahassee, FL**
B.S. Physics, Summa Cum Laude — *2020 (Fall) – 2023 (Spring)*
Honors Thesis: Study of Displaced Vertex Tagging with the CMS Experiment
Advisor: Prof. Ted Kolberg

Emory University — **Atlanta, GA**
Foundational Coursework — *2018 (Fall) – 2020 (Spring)*

Publications

5. V. Fondement, **E. Todd**, and I. Jovanovic, "Measurement of ^{78m}Br $2^- \rightarrow 1^+$ Transition Half-Life by Self-Coincidence Analysis in an Active Target $LaBr_3$(Ce) Detector," *in preparation*, 2026
4. **E. Todd**, V. Fondement, and I. Jovanovic, "Machine Learning-Enabled Pulse Shape Discrimination in YAP:Ce," *under review*,
3. C. Graham, **E. Todd**, C. Wilson, et al., "Light Output-Dependent Integration Gating for Optimized Pulse Shape Discrimination," *under review*,
2. CMS HGCAL Collaboration, "Leakage current of high-fluence neutron-irradiated 8" silicon sensors for the CMS Endcap Calorimeter Upgrade," *under review*, 2025. [Online]. Available: https://arxiv.org/abs/2510.17280
1. K. Yocum, H. Smith, **E. Todd**, et al., "Millimeter/Submillimeter Spectroscopic Detection of Desorbed Ices: A New Technique in Laboratory Astrochemistry," *The Journal of Physical Chemistry A*, vol. 123, no. 40, pp. 8702–8708, 2019, ISSN: 10.1021/acs.jpca.9b04587

Research Experience

Applied Nuclear Science Group, University of Michigan — **Ann Arbor, MI**
Graduate Student Researcher — *2024 (Summer) – Present*

Particle Identification in YAP:Ce via Machine Learning

- Compared α and γ-ray discrimination capabilities of charge comparison/shape indicator to several machine learning methods: random forests, gradient boosting, XGBoost, and multilayer perceptrons.
- Quantifed the performance of each method using receiver operating characteristic curves, and demonstrated the substantial advantage of machine learning over traditional methods.
- Applied the XGBoost model to proof-of-concept mixed field radiation environments consisting of ^{241}Am and ^{137}Cs or ^{241}Am and ^{60}Co, and demonstrated that known spectral features of ^{241}Am, ^{137}Cs, and ^{60}Co are correctly classified.

Investigation of the Ionization Yield of Germanium at 254 eV_{nr}

Collaborator: Dr. Alexander Kavner, University of Zurich
- Jointly conducted an experimental campaign utilizing the spallation neutron source at the Paul Scherrer Institute to re-measure the energy of the final gamma-ray emitted in the ^{72}Ge(n, γ)^{73}Ge thermal neutron capture reaction, which is necessary to validate a recent ANSG measurement of the ionization yield of Ge.
- Utilized a state-of-the-art, position sensitive CZT detector to conduct precision measurements of the final state gamma-ray energy at the Ohio State Nuclear Reactor Laboratory thermal neutron beam facility.

Measurement of the ^{37}Cl(a, n)^{40}K Cross Section

Collaborator: Dr. Daniel Santiago-Gonzalez, Argonne National Laboratory

- Currently working on using data from the Multi-Sampling Ionization Chamber (MUSIC) experiment at ANL to produce the first measurement of the ^{37}Cl(a, n)^{40}K cross section.
- Validated stopping power tables for ^{37}Cl in helium gas up to 2.5 MeV/u by comparing results from LISE^{++} and Geant4 calculations to calibrated measurements with a silicon diode.
- Developing an event reconstruction algorithm for MUSIC which performs offline time synchronization of four digitizers via analysis of the time structure of the ^{37}Cl beam.

CMS Group, Florida State University — **Tallahassee, FL**

Undergraduate Student Researcher — *2021 – 2023* (Fall – Summer)

Search for BSM Long-lived Particles

Advisor: Prof. Ted Kolberg

- Utilized deep neural network machine learning to search for beyond the Standard Model long-lived particles in data from the CMS Experiment.
- Trained ML models on Monte Carlo data with varying signal (mass, lifetime) and background (t, b physics) to verify their ability to identify displaced vertices independently of the particular physics.
- Investigated the most effective parameters of ML models and verified physics-based expectations.

Silicon Sensor Quality Control

Advisor: Prof. Rachel Yohay

- Conducted process quality control (PQC) testing for the CMS High Granularity Calorimeter (HGCAL) to monitor the quality and stability of the silicon sensor production process.
- Analyzed and presented PQC results to the CMS HGCAL sensor testing group.
- Helped create and implement a standardized testing procedure at FSU.

WW Astrochemistry Group, Emory University — **Atlanta, GA**

Undergraduate Student Researcher — *2018 – 2020* (Fall – Fall)

Advisor: Prof. Susanna Widicus Weaver

- Performed millimeter/submillimeter spectroscopic measurements of desorbed astrophysical ice analogs.
- Conducted proof-of-concept millimeter/submillimeter spectroscopic measurements of H_2O and D_2O binding energy, sublimation enthalpy, and sublimation entropy.
- Analyzed broadband molecular line surveys using the GOBASIC software package.

Teaching Experience

University of Michigan — **Ann Arbor, MI**

Lead Graduate Student Instructor — *2025 - 2026* (Fall - Winter)

- Oversee 30 graduate student instructors teaching general physics laboratories with a total enrollment of 1800 undergraduate students.
- Edit and maintain the laboratory manuals and Jupyter notebook worksheets used in the course.
- Manage the course information and organization via Canvas.
- Arbitrate student and graduate student instructor concerns.

University of Michigan — **Ann Arbor, MI**

Graduate Student Instructor — *2023 – 2025* (Fall – Spring)

- 2025 (Spring) Physics 251, Fundamental Physics for the Life Sciences II Lab
- 2024 (Fall) Physics 121, Physics for Architects Lab
- 2023 – 2024 (Fall – Winter) Physics 241, General Physics II Lab

Florida State University — **Tallahassee, FL**
Physics Learning Assistant — *2022* *Spring*

PHY 2049C, General Physics B

Emory University — **Atlanta, GA**
Laboratory Teaching Assistant — *2019* *Fall*

CHEM 150L, Structure and Properties Lab

Honors, Awards, and Grants

- **2025** **Fall** : NSUF Rapid Turnaround Experiment Grant — *Awarded to conduct precision measurements of Ge gamma-ray cascades at the Ohio State University Nuclear Reactor Laboratory.*
- **2025** **Fall** : Rackham Graduate Student Research Grant — *Awarded to conduct research at the Paul Scherrer Institute on the ionization yield of high-purity germanium detectors.*
- **2023** **Spring** : Phi Beta Kappa Society
- **2023** **Spring** : Joseph Lanutti Undergraduate Research Award — *Awarded for third place in the FSU physics departmental poster presentation contest.*
- **2022** **Fall** : Evelyn and John Baugh Research Presentation Scholarship — *Awarded to travel and present research at SESAPS.*
- **2022** **Spring** : Anna Runyan Undergraduate Endowment — *Awarded for outstanding academic success in physics.*

Skills

Programming: C++, Python, Nix, C

Software:

Nuclear Measurements:	ROOT, Geant4, CoMPASS, wavedump, LISE^{++}, SRIM
Machine Learning:	Scikit-learn, XGBoost, Tensorflow
Version Control:	Git, Nix Flakes
Build Systems:	Nix, Make, CMake
Scientific Presentation:	LaTeX/Beamer
Firmware:	coreboot/edk2, Chromium Embedded Controller

Hardware: CAEN digitizers, soldering, electronic testing equipment (*i.e.,* oscilloscopes)

Presentations

4. **E. Todd**, V. Fondement, and I. Jovanovic, "Particle Identification in Fast Inorganic Scintillators via Machine Learning," *Fall Meeting of the Eastern Great Lakes Section of the American Physical Society, Ypsilanti, MI*, 2025

3. **E. Todd** and I. Jovanovic, "Investigation of Anomalously High Ionization Produced by Low Energy Nuclear Recoils in Germanium (poster)," *DOE NNSA Office of Defense Nuclear Nonproliferation R&D University Program Review, Gainesville, FL*, 2025

2. **E. Todd**, A. A. Kadhim, N. Bower, et al., "Process Quality Control for HGCAL at Florida State University (poster)," *89th Annual Meeting of the Southeastern Section of the American Physical Society, Oxford, MS*, 2022

1. **E. Todd**, K. Yocum, P. Gerakines, et al., "A Novel Use of Rotational Spectroscopy for Studying Characteristics of Desorbed Interstellar Ice Analogs (poster)," *Emory SURE 2019 Symposium, Atlanta, GA*, 2019